

Integrate acoustic emission sensors along the infusion tubing that analyze pressure wave signatures characteristic of occlusion events

Executive Technology Brief

Domain: fluidics_hydraulic

Technology Maturity: ENGINEERING_DEFINITION

Transfer Posture: Sponsored Validation

Problem: Documented recurring problem 'downstream occlusion detection failure' for infusion pumps: 100 MAUDE adverse-event records retrieved live at benchmark time (2007–2032). Event narratives: 'custody in retrieval log'. Literature context: 10 EuropePMC records on infusion-pump occlusion alarms. MAUDE LIMITS STAMPED INLINE: incidence UNKNOWN, causation UNVERIFIED, voluntary reports (Art. XXI.5).

Mechanism: Industrial fluid dynamics monitoring system using pressure wave pattern recognition

Architecture: PROPOSED architecture (no physical system exists): flow path / lumen architecture; pressure regulation element; sensing / feedback element (if closed-loop); termination interfaces (patient / reservoir) — realizing: Integrate acoustic emission sensors along the infusion tubing that analyze pressure wave signatures characteristic of occlusion events

Governing Model: Fluid mechanics / hydraulics governing relations proposed for this invention; SYMBOLIC ONLY until inputs are sourced

Key Equations:

FLUID-001: Hagen-Poiseuille (laminar pipe flow). Expression: $Q = (\pi * r^4 * \Delta P) / (8 * \mu * L)$.

Applicability to this invention: APPLICABLE; holds under: Newtonian fluid, laminar regime, rigid circular lumen, fully developed flow. Assumptions: CSF approximated as Newtonian (valid at shunt flow rates); no wall compliance effects. Engaged variables: ΔP , μ .

FLUID-002: Reynolds number. Expression: $Re = (\rho * v * D) / \mu$. Applicability to this invention:

APPLICABLE; holds under: circular section; single-phase flow. Assumptions: transition criterion $Re \sim 2300$ is a MODEL_DERIVED engineering rule of thumb. Engaged variables: μ .

FLUID-003: Orifice flow (sharp-edged restriction). Expression: $Q = C_d * A * \sqrt{2 * \Delta P / \rho}$.

Applicability to this invention: APPLICABLE; holds under: turbulent orifice flow; C_d treated as MODEL_DERIVED until measured. Assumptions: $C_d \sim 0.6$ - 0.65 is an engineering prior, not a measurement. Engaged variables: ΔP .

Kill Condition: killer experiment fails: falsification_test_from_candidate (decision scores: EIG 0.44573539382492045, EIG-per-cost 0.44573539382492045; the falsification test carries the pre-registered pass/fail before any transfer)

Current State: ENGINEERING_DEFINITION. No physical prototype. No clinical validation.

Disclosure: This dossier distinguishes established evidence from proposals and unresolved questions.